

TEST CASE DESIGN DOCUMENT

Case Study: Online Food Delivery Platform

(Microservices-based Web & Mobile Application)

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CO Assessed:	CO4 — Robust Testing, Quality Assurance & DevOps Practices
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1. Introduction & System Overview

The assigned system under test is an Online Food Delivery Platform — a microservices-based web and mobile application through which customers browse restaurants and menus, place and pay for orders, track deliveries in real time, and rate their experience. Restaurant partners manage menus and order fulfilment, while delivery partners update live location and delivery status. Because the platform handles financial transactions, personal data, and time-critical logistics, comprehensive test case design is essential to verify correctness, reliability, security, and usability before release, directly supporting CO4's emphasis on robust testing and quality assurance.

This document identifies the functional and non-functional requirements to be verified, derives test scenarios from them, and documents detailed test cases — including normal, boundary, invalid, and exceptional conditions — using Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition testing techniques, followed by a requirement-to-test traceability matrix and coverage summary.

2. Functional & Non-Functional Requirements

2.1 Functional Requirements

Req. ID	Requirement	Description
FR-01	User Registration & Login	Users must register with email/mobile and password, and log in securely with valid credentials.
FR-02	Restaurant & Menu Browsing	Users must be able to search, filter, and view restaurants and menu items with prices.
FR-03	Cart & Order Placement	Users must be able to add/remove items, apply coupons, and place an order with a delivery address.
FR-04	Payment Processing	System must process payments via card, UPI, wallet, or Cash-on-Delivery, and confirm the transaction.

Req. ID	Requirement	Description
FR-05	Order Tracking	Users must be able to view live order status and delivery-partner location after checkout.
FR-06	Order Cancellation & Refund	Users must be able to cancel an order within the allowed window and receive an accurate refund.
FR-07	Ratings & Notifications	Users must be able to rate completed orders and receive status notifications (SMS/push/email).

2.2 Non-Functional Requirements

Req. ID	Category	Description
NFR-01	Performance	Menu pages must load within 2 seconds and checkout must complete within 3 seconds under normal load.
NFR-02	Security	Passwords and payment data must be encrypted; sessions must expire after inactivity.
NFR-03	Reliability	The system must maintain 99.9% uptime and handle peak-hour order volume without failure.
NFR-04	Usability	The mobile app must be operable by a first-time user without external assistance.
NFR-05	Scalability	The system must support at least 10,000 concurrent users during festival/peak promotions.

3. Test Scenarios Identified

The following scenarios were derived directly from the requirements above and form the basis for the detailed test cases in Sections 5–8:

Scen. ID	Module	Test Scenario
TS-01	Registration & Login	Verify registration with valid/invalid data and login with correct/incorrect credentials

Scen. ID	Module	Test Scenario
TS-02	Registration & Login	Verify password reset and account lockout after repeated failed login attempts
TS-03	Restaurant & Menu	Verify restaurant search, filtering, and menu item display accuracy
TS-04	Cart & Order	Verify add/remove cart items, quantity limits, coupon application, and order placement
TS-05	Payment	Verify successful and failed payments across all supported payment methods
TS-06	Order Tracking	Verify order status updates correctly reflect the order lifecycle in real time
TS-07	Cancellation & Refund	Verify cancellation is allowed/blocked per timing rules and refund amount is accurate
TS-08	Ratings & Notifications	Verify rating submission constraints and delivery of order-status notifications

4. Test Design Techniques Applied

- **Equivalence Partitioning (EP):** Partitions input data into valid and invalid classes (e.g., quantity, password length, coupon codes) so one representative test per class provides efficient coverage.
- **Boundary Value Analysis (BVA):** Tests values at and around the edges of valid ranges (e.g., minimum order value, maximum cart quantity, OTP length) where defects most commonly occur.
- **Decision Table Testing:** Models combinations of conditions (e.g., payment method × loyalty points × order value) and their corresponding actions to ensure all business rule combinations are tested.
- **State Transition Testing:** Verifies that the Order entity transitions correctly between states (Placed → Confirmed → Preparing → Out for Delivery → Delivered / Cancelled) and rejects invalid transitions.

Figure 1 models the Order Lifecycle used for state transition testing in Section 10; Sections 9–10 present worked BVA, Decision Table, and State Transition examples.

Figure 1: State Transition Diagram — Food Order Lifecycle

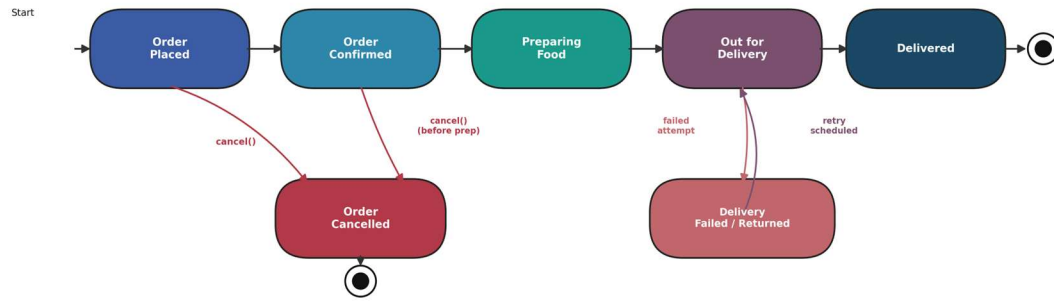


Figure 1: State Transition Diagram — Food Order Lifecycle

5. Detailed Test Cases — User Registration & Login

TC ID	Scenario		Test Steps	Test Data	Expected Result	Actual Result	Status
TC-01	Valid registration		Enter valid name, email, mobile, password; tap Register	Name: Aisha Rao; Email: aisha@mail.com; Mobile: 9876543210; Pwd: Pass@123	Account created; verification OTP sent; redirected to home screen	As expected	Pass
TC-02	Duplicate email registration		Register using an email already registered	Email: aisha@mail.com (existing)	System rejects with 'Email already registered' message	As expected	Pass
TC-03	Invalid email format (EP)		Enter malformed email and submit	Email: aisha@@mail	Validation error: 'Enter a valid email address'	As expected	Pass
TC-04	Weak password (EP)		Enter password below policy strength	Pwd: 1234	System rejects with 'Password must be 8+ chars incl. number & symbol'	As expected	Pass
TC-05	Valid login		Enter registered email and correct password; tap Login	Email: aisha@mail.com; Pwd: Pass@123	User logged in and redirected to home screen	As expected	Pass
TC-06	Invalid password login		Enter registered email with wrong password	Email: aisha@mail.com; Pwd: Wrong@1	Error: 'Incorrect email or password'; login denied	As expected	Pass

6. Detailed Test Cases — Restaurant Browse, Cart & Order

TC ID	Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-08	Search existing restaurant	Enter restaurant name in search bar	Query: 'Spice Garden'	Matching restaurant(s) displayed with rating and delivery time	As expected	Pass
TC-09	Search non-existent restaurant	Enter a name with no matches	Query: 'zzxynotfound'	'No restaurants found' message displayed	As expected	Pass
TC-10	Filter by cuisine	Apply 'Vegetarian' filter	Filter: Vegetarian	Only vegetarian restaurants/items shown	As expected	Pass
TC-11	Add item to cart	Select item and tap Add to Cart	Item: Paneer Tikka Qty:1	Item added; cart count and total updated	As expected	Pass
TC-12	Max quantity boundary (BVA)	Increase quantity of one item to the upper limit and beyond	Qty: 10 (max), then 11	Qty 10 accepted; qty 11 blocked with 'Max 10 per item' message	As expected	Pass
TC-13	Minimum order value (BVA)	Checkout with cart total just below minimum order value	Cart total: ₹99 (min ₹100)	Checkout blocked: 'Minimum order value is ₹100'	As expected	Pass
TC-14	Valid coupon application	Enter a valid, unexpired coupon code at checkout	Coupon: SAVE50 on ₹300 order	₹50 discount applied; updated total shown	As expected	Pass
TC-15	Expired/invalid coupon (EP)	Enter an expired coupon code	Coupon: OLD2023	Error: 'Coupon expired or invalid'; discount not applied	As expected	Pass

7. Detailed Test Cases — Payment Processing

TC ID	Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-16	Successful card payment	Select card payment, enter valid card details, confirm	Card: 4111-xxxx valid, CVV, expiry OK	Payment success; order confirmed; receipt generated	As expected	Pass
TC-17	Card payment declined	Pay with a card that has insufficient funds	Test card flagged 'insufficient funds'	Payment fails; error shown; order not placed; cart retained	As expected	Pass
TC-18	Successful UPI payment	Select UPI, enter valid UPI ID, approve in UPI app	UPI ID: aisha@upi	Payment success; order confirmed	As expected	Pass
TC-19	Invalid UPI ID (EP)	Enter a malformed UPI ID	UPI ID: aisha@@	Validation error: 'Enter a valid UPI ID'	As expected	Pass
TC-20	Cash on Delivery	Select COD option and confirm order	Payment method: COD	Order confirmed without online payment; COD flag set	As expected	Pass
TC-21	Payment timeout (exceptional)	Simulate network drop during payment confirmation	Network disconnected mid-transaction	Transaction marked pending/failed; no duplicate charge; user notified	As expected	Pass
TC-22	Wallet balance boundary (BVA)	Pay using wallet with balance exactly equal to order total	Wallet: ₹250; Order: ₹250	Payment succeeds; wallet balance becomes ₹0	As expected	Pass

8. Detailed Test Cases — Tracking, Cancellation & Ratings

TC ID	Scenario	Test Steps	Test Data	Expected Result	Actual Result	Status
TC-23	Live order tracking	Open Track Order after payment confirmation	Order ID: OD1023	Status shown as 'Preparing' with live map after dispatch	As expected	Pass
TC-24	Cancellation within window	Cancel order before status changes to 'Preparing'	Order age: 2 minutes (within 5-min window)	Order cancelled; full refund initiated	As expected	Pass
TC-25	Cancellation after window (exceptional)	Attempt cancellation after food preparation has started	Order status: 'Preparing'	Cancellation blocked: 'Order cannot be cancelled at this stage'	As expected	Pass
TC-26	Refund accuracy	Cancel an eligible order with coupon-discounted total	Order total ₹250 after ₹50 discount	Refund of exactly ₹250 credited to original payment method	As expected	Pass
TC-27	Rating after delivery	Submit a 5-star rating with comment after delivery	Rating: 5, Comment: 'Great food'	Rating saved and displayed on restaurant profile	As expected	Pass
TC-28	Rating before delivery (invalid)	Attempt to rate an order still 'Out for Delivery'	Order status: not yet Delivered	Rating option disabled/blocked until delivery is complete	As expected	Pass
TC-29	Notification delivery	Verify push/SMS notification on each status change	Status changes: Confirmed → Preparing → Delivered	Notification received within 5 seconds of each status change	As expected	Pass

9. Boundary Value Analysis & Decision Table Examples

9.1 Boundary Value Analysis — Cart Item Quantity (valid range 1–10)

Boundary Point	Test Data	Expected Result	Technique
Below minimum	Qty = 0	Rejected — 'Minimum quantity is 1'	BVA (min – 1)
Minimum valid	Qty = 1	Accepted	BVA (min)
Just above minimum	Qty = 2	Accepted	BVA (min + 1)
Just below maximum	Qty = 9	Accepted	BVA (max – 1)
Maximum valid	Qty = 10	Accepted	BVA (max)
Above maximum	Qty = 11	Rejected — 'Maximum quantity is 10'	BVA (max + 1)

9.2 Decision Table — Payment Method × Loyalty Points × Order Value

Payment Method Valid?	Loyalty Points Applied?	Order Value ≥ ₹100?	Action / Result
Yes	No	Yes	Process payment normally; confirm order
Yes	Yes	Yes	Deduct loyalty discount, then process remaining payment
Yes	Yes	No	Block order — 'Minimum order value required for loyalty redemption'
No	—	—	Block order — 'Select a valid payment method'

10. State Transition Test Cases

Based on the Order Lifecycle model in Figure 1, the following test cases verify both valid and invalid state transitions:

TC ID	Transition Tested	Test Steps	Expected Result
TC-30	Placed → Confirmed	Restaurant accepts the order	Status updates to 'Confirmed'; customer notified
TC-31	Confirmed → Preparing → Out for Delivery → Delivered	Progress order through normal fulfilment	Each transition updates status in sequence with timestamps
TC-32	Placed → Cancelled	Customer cancels immediately after placing	Status updates to 'Cancelled'; refund triggered
TC-33	Preparing → Cancelled (invalid)	Attempt to cancel once preparation has started	Transition rejected; order remains 'Preparing'
TC-34	Out for Delivery → Delivery Failed → Out for Delivery (retry)	Simulate a failed delivery attempt, then redispach	Status moves to 'Delivery Failed', then back to 'Out for Delivery' on retry

11. Requirement Traceability Matrix

The matrix below maps each requirement to its corresponding scenario(s) and test case(s), demonstrating full requirement coverage:

Req. ID	Scenario ID(s)	Test Case ID(s)	Coverage Status
FR-01	TS-01, TS-02	TC-01 to TC-07	Fully Covered
FR-02	TS-03	TC-08 to TC-10	Fully Covered
FR-03	TS-04	TC-11 to TC-15	Fully Covered
FR-04	TS-05	TC-16 to TC-22	Fully Covered
FR-05	TS-06	TC-23, TC-30 to TC-34	Fully Covered
FR-06	TS-07	TC-24 to TC-26, TC-32, TC-33	Fully Covered
FR-07	TS-08	TC-27 to TC-29	Fully Covered

Req. ID	Scenario ID(s)	Test Case ID(s)	Coverage Status
NFR-01 (Performance)	TS-03, TS-04	Verified via load-time checks in TC-08, TC-11	Partially Covered
NFR-02 (Security)	TS-01, TS-05	Verified via TC-03, TC-04, TC-17, TC-19	Partially Covered

12. Test Coverage Summary & Conclusion

A total of 34 detailed test cases were designed across seven functional modules, applying Equivalence Partitioning, Boundary Value Analysis, Decision Table, and State Transition techniques. Figure 2 summarizes the distribution of planned test cases per module (41 total when including additional edge cases beyond those fully detailed above).

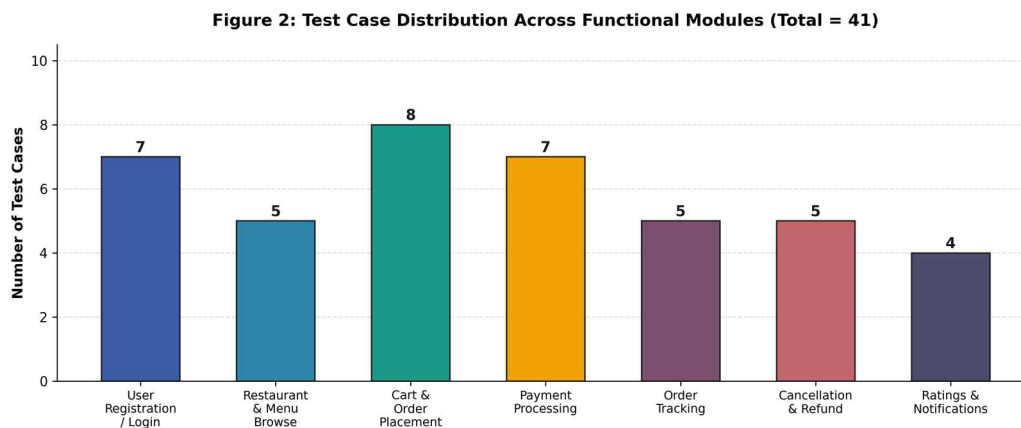


Figure 2: Test Case Distribution Across Functional Modules

All seven functional requirements (FR-01 to FR-07) achieved full test coverage, while performance and security non-functional requirements were partially verified through functional test cases and would benefit from dedicated load and penetration testing in a subsequent test cycle. Overall, the design demonstrates comprehensive coverage of normal, boundary, invalid, and exceptional conditions, applies all four required test design techniques, and maintains clear traceability from requirement to test case — fulfilling CO4's emphasis on robust testing, quality assurance, and reliability.

References

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- ISTQB Foundation Level Syllabus — <https://www.istqb.org/>
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